Straightening Laws in Algebraic Complexity

Anakin Dey

What is Algebraic Complexity?

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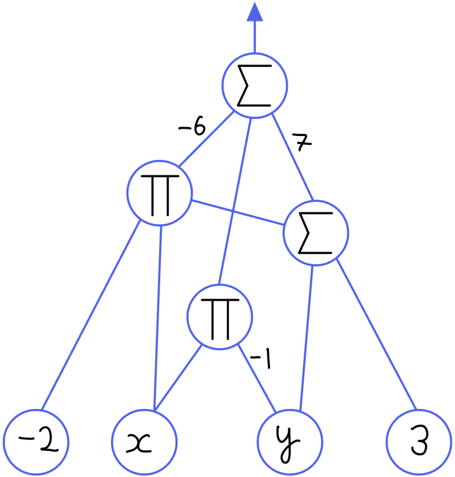
Can we find *explicit* polynomials which are hard to compute?

Algebraic Circuits



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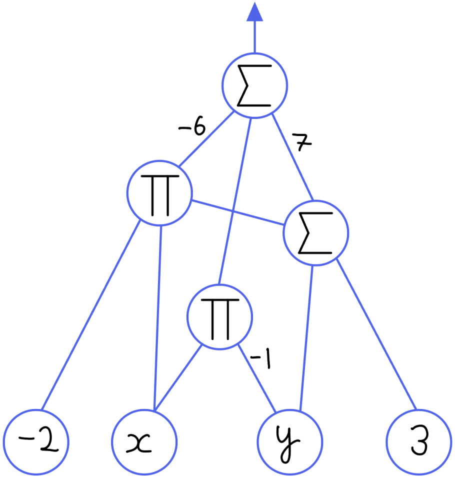

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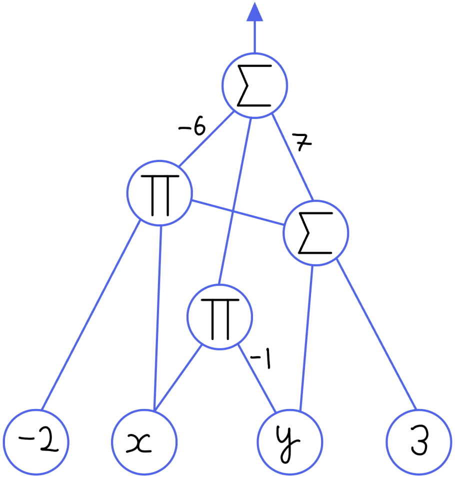

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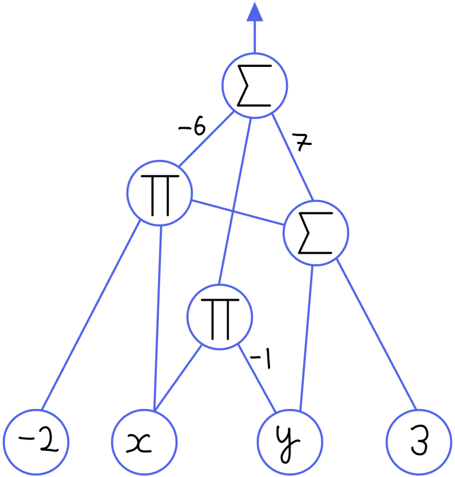

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- *Size:* The number of edges
- *Depth:* The length of the longest input-output path

Horner's Rule

Let $f(x) = \sum_{i=0}^{d} a_i x^i \in \mathbb{F}[x]$ be a univariate polynomial.

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- This gives a circuit using exactly d additions and d multiplications.
- The number of additions is optimal [Ost54].
- The number of multiplications is optimal [Pan66].

Hard Polynomials Exist

Proposition

Let $\mathbb{F}$ be a field of characteristic zero. Then "most" n-variate polynomials of degree $\leq d$ require circuits of size $N := \binom{n+d}{d}$.

Sketch

- Count the number of unlabeled circuits of size s.
- For a circuit of size s, labeling the edges gives a map $\Phi \colon \mathbb{F}^s \to \mathbb{F}^N$
- $\dim \operatorname{im} \Phi \leq s$, and there are only finitely many unlabeled circuits. Thus, the dimension over all possible unlabelled circuits is at most s.
- For the image to be Zariski-dense, we must have that $s \geq N$.

Determinant versus Permanent

Conjecture

Let $\mathbb{F}$ be a field, $\mathrm{char}(\mathbb{F}) \neq 2$, $X = (x_{i,j})$ be $n \times n$ matrix of variables.

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Suppose that A is an $m \times m$ matrix with entries in $\mathbb{F}[X]_{\leq 1}$ such that

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Then m must be superpolynomial in n.

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Definition

We say a circuit *border computes* $g(\overline{x})$ over $\mathbb{F}$ if it computes

$$g(\overline{x}) + \varepsilon^q \tilde{g}(\overline{x}, \varepsilon) \in \mathbb{F}[[\varepsilon]][\overline{x}], \qquad q \geq 1.$$

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We cannot set $\varepsilon = 0$ since we may divide by ε.

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What is the power of border computation?

Is it strictly necessary? Can we prove something in the border setting and *deborder* the result to get an exact computation?

Border Waring Rank

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Definition

The *Waring Rank*, $\mathrm{WR}(f)$, of a degree d form $f(\overline{x})$ over $\mathbb{C}$ is the minimum r such that there exist linear forms $\ell_1(\overline{x}), \ldots, \ell_r(\overline{x})$ such that

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Thus, the *Border Waring Rank*, $\overline{\mathrm{WR}}(f)$, is equal to 2.
In other words, $\{ f(x,y) \in \mathbb{C}[x,y]_{\leq d} \mid \mathrm{WR}(f) = 2 \}$ is not Zariski-closed.

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Suppose $f \in \langle g_1, \ldots, g_k \rangle$. How does the complexity of f compare to the complexity of the generators $g_1, \ldots, g_k$?

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How can we compare the *relative* complexity of two polynomials?

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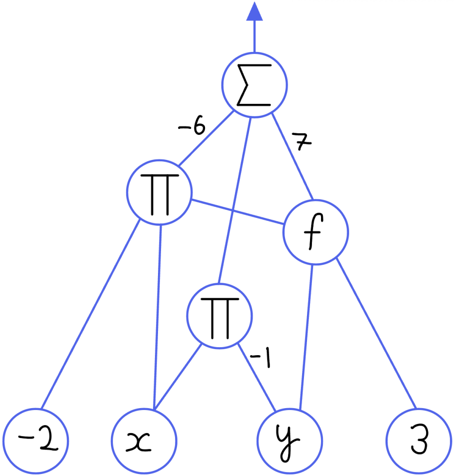

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How can we compare the *relative* complexity of two polynomials?

We can allow a fixed polynomial $f(\overline{x})$ to be computed with unit cost by allowing it to function as its own gate.

Principal Ideals

Example

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Question

Suppose $f \in \langle g \rangle$. Does g have a small f-oracle circuit?

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If g is a factor of f, $\mathrm{size}(g) \leq \mathrm{poly}(\mathrm{size}(f), \deg(f))$.

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This is open for *general* polynomials $g_1(\overline{x}), g_2(\overline{x})$ [Gro20].

This is open even if we ask that $\{\, g_1, g_2 \,\}$ are a Gröbner basis or form a regular sequence.

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Adding $\det_n$ to the set of generators remedies this issue.

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Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I^{\text{det}}_{n,m,r}$ be the *determinantal ideal* generated by the $r \times r$ minors of X.

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- However, these circuits are *not* constant-depth.
- The oracle allows *constant depth* computation of the determinant.

Closure Results in Determinantal Ideals

Theorem ([AF22, Theorem 1.1])

Let $f \in I^{\det}_{n,m,r}$ be a nonzero polynomial. Then there is a depth-three f-oracle circuit of size $O\left(n^2m^2\right)$ that *border computes* the $t \times t$ determinant for some $t = \operatorname{poly}\left(r^{1/3}\right)$.

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Closure Results in Determinantal Ideals

Theorem ([DG25, Theorem 1.5])

Let $f \in I^{\det}_{n,m,r}$ be a nonzero polynomial. Then there is a depth-three f-oracle circuit of size $\mathrm{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = \mathrm{poly}\left(r^{1/3}\right)$.

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Main Tools:

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Main Tools:

- We use *Straightening Laws* from Invariant Theory to express $f(X)$ in a standard basis indexed by Young tableau, and leverage the combinatorics to talk about specific terms.
- To get a circuit for a specific basis term, we use *Interpolation* as well as a new application of the *Isolation Lemma*.

Dealing with Cancellations

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- Working with the monomial basis is hard since determinants have exponentially many terms.
- Thus, we will work with a different basis for $I^{\det}_{n,m,r}$ which is described combinatorially with Young tabelau.

Bideterminants

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$$(S \mid T) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 4 & 5 \\ \hline 3 & 6 \\ \cline{1-2} 4 \\ \cline{1-1} \end{array} \;\middle|\; \begin{array}{|c|c|c|c|} \hline 1 & 3 & 5 & 6 \\ \hline 2 & 7 \\ \cline{1-2} 3 \\ \cline{1-1} \end{array} \right).$$

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Then the *bideterminant* $(S \mid T)(X)$ is given by

$$(S \mid T)(X) = \begin{vmatrix} x_{1,1} & x_{1,3} & x_{1,5} & x_{1,6} \\ x_{2,1} & x_{2,3} & x_{2,5} & x_{2,6} \\ x_{4,1} & x_{4,3} & x_{4,5} & x_{4,6} \\ x_{5,1} & x_{5,3} & x_{5,5} & x_{5,6} \end{vmatrix} \begin{vmatrix} x_{3,2} & x_{3,7} \\ x_{6,2} & x_{6,7} \end{vmatrix} \left(x_{4,3} \right).$$

A Natural Generating Set for Determinantal Ideals

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Thus, the bideterminants span $\mathbb{F}[X]$.

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Theorem ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3])

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Theorem ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3])

Let $(S \mid T)(X)$ be a bideterminant of shape σ. Then $(S \mid T)(X)$ can be uniquely expressed as a linear combination

$$(S \mid T)(X) = \sum_{(A|B)} c_{A,B}(A \mid B)(X),$$

where A and B are *conjugate semistandard*: strictly increasing along rows, weakling increasing down columns.

A Natural Generating Set for Determinantal Ideals

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Proposition ([BC03, Corollary 1.7])

The ideal $I^{\det}_{n,m,r}$ has as basis the conjugate semistandard bideterminants of shape σ whose first row has length at least r.

Shuffle Products

$$\left(\begin{array}{|c|c|c|c|c|c|}\hline i_1 \;\bullet & \cdots & i_p \;\bullet & i_{p+1} & \cdots & i_s \\ \hline \ell_1 \;\bullet & \cdots & \ell_q \;\bullet & \ell_{q+1} & \cdots & \ell_t \\ \hline\end{array} \quad\middle|\quad \begin{array}{|c|c|c|}\hline j_1 & \cdots & j_s \\ \hline m_1 & \cdots & m_t \\ \hline\end{array}\right)$$

$$= \sum_{\sigma} \operatorname{sgn}(\sigma) \left(\begin{array}{|c|c|c|c|c|c|}\hline \sigma(i_1) & \cdots & \sigma(i_p) & i_{p+1} & \cdots & i_s \\ \hline \sigma(\ell_1) & \cdots & \sigma(\ell_q) & \ell_{q+1} & \cdots & \ell_t \\ \hline\end{array} \quad\middle|\quad \begin{array}{|c|c|c|}\hline j_1 & \cdots & j_s \\ \hline m_1 & \cdots & m_t \\ \hline\end{array}\right)$$

Where the summation is over all permutations σ such that $\sigma(i_1) < \cdots < \sigma(i_p)$ and $\sigma(\ell_1) < \cdots < \sigma(\ell_q)$.

Laplace Expansion

$$\left(\begin{array}{|c|c|c|c|c|c|}\hline i_1^{\bullet} & \cdots & i_p^{\bullet} & i_{p+1} & \cdots & i_s \\ \hline \ell_1^{\bullet} & \cdots & \ell_q^{\bullet} & \ell_{q+1} & \cdots & \ell_t \\ \hline \end{array} \;\middle|\; \begin{array}{|c|c|c|}\hline j_1 & \cdots & j_s \\ \hline m_1 & \cdots & m_t \\ \hline \end{array}\right)$$

$$= \sum_{\tau,\mu} \operatorname{sgn}(\tau)\operatorname{sgn}(\mu) \left(\begin{array}{|c|c|c|c|c|c|}\hline i_1 & \cdots & i_p & \ell_1 & \cdots & \ell_q \\ \hline i_{p+1} & \cdots & i_s \\ \cline{1-3} \ell_{q+1} & \cdots & \ell_t \\ \cline{1-3} \end{array} \;\middle|\; \begin{array}{|c|c|c|c|c|c|}\hline \tau(j_1) & \cdots & \tau(j_p) & \mu(m_1) & \cdots & \mu(m_q) \\ \hline \tau(j_{p+1}) & \cdots & \tau(j_s) \\ \cline{1-3} \mu(m_{q+1}) & \cdots & \mu(m_t) \\ \cline{1-3} \end{array}\right)$$

Where the summation is over all permutations σ such that $\sigma(i_1) < \cdots < \sigma(i_p)$ and $\sigma(\ell_1) < \cdots < \sigma(\ell_q)$.

Example Straightening

Consider the following nonstandard tableau

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline 2 \\ \cline{1-1} \end{array} \;\middle|\; \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 \\ \cline{1-1} \end{array} \right)$$

This is not conjugate semistandard in the first two rows, so we focus on straightening these first two rows.

Example Straightening

Shuffle:

$$\left(\begin{ytableau} 2 & 3 \\ 1 & 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \end{ytableau}\right) = \left(\begin{ytableau} 2 & 3 \\ 1 & 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \end{ytableau}\right) - \left(\begin{ytableau} 1 & 3 \\ 2 & 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \end{ytableau}\right) + \left(\begin{ytableau} 1 & 2 \\ 3 & 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \end{ytableau}\right)$$

Laplace:

$$\begin{aligned}\left(\begin{ytableau} 2 & 3 \\ 1 & 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \end{ytableau}\right) &= \left(\begin{ytableau} 1 & 2 & 3 \\ 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 1 & 2 \\ 3 \end{ytableau}\right) - \left(\begin{ytableau} 1 & 2 & 3 \\ 4 \end{ytableau} \;\middle|\; \begin{ytableau} 3 & 1 & 2 \\ 1 \end{ytableau}\right) \\ &= -\left(\begin{ytableau} 1 & 2 & 3 \\ 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 & 3 \\ 1 \end{ytableau}\right)\end{aligned}$$

We notice that our unstraightened rows appear in the first expression so that we can solve for it.

Example Straightening

$$\left(\begin{ytableau} 2 & 3 \\ 1 & 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \end{ytableau}\right) = \left(\begin{ytableau} 1 & 3 \\ 2 & 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \end{ytableau}\right) - \left(\begin{ytableau} 1 & 2 \\ 3 & 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \end{ytableau}\right) - \left(\begin{ytableau} 1 & 2 & 3 \\ 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 & 3 \\ 1 \end{ytableau}\right)$$

Thus, we get that

$$\left(\begin{ytableau} 2 & 3 \\ 1 & 4 \\ 2 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \\ 1 \end{ytableau}\right) = \left(\begin{ytableau} 1 & 3 \\ 2 & 4 \\ 2 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \\ 1 \end{ytableau}\right) - \left(\begin{ytableau} 1 & 2 \\ 3 & 4 \\ 2 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \\ 1 \end{ytableau}\right) - \left(\begin{ytableau} 1 & 2 & 3 \\ 4 \\ 2 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 & 3 \\ 1 \\ 1 \end{ytableau}\right)$$

The first term is conjugate semistandard, but not the next two terms.

All terms that appear have the same content and they are either the same shape with lexicographically smaller filling or they have a longer shape.

Main Theorem

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I^{\det}_{n,n,n/2}$ be a nonzero polynomial. Then there is a constant depth f-oracle circuit of size $\mathrm{poly}(n)$ that computes the $t \times t$ determinant for some $t = \mathrm{poly}(n)$.

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Theorem ([DG25, Theorem 1.5])

Let $f \in I^{\det}_{n,m,r}$ be a nonzero polynomial. Then there is a depth-three f-oracle circuit of size $\mathrm{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = \mathrm{poly}\left(r^{1/3}\right)$.

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Let $f(X) \in I^{\mathrm{det}}_{n,m,r}$ be a nonzero polynomial of degree d. Then, there exists a depth-three f-oracle circuit of size $O\left(n^2m^2d^3(n^3+m^3)\right)$ computing $(K_\sigma \mid K_\sigma)(X)$, where $\sigma_1 \geq r$ and $|\sigma| \leq d$.

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Let $\sigma = (4, 2, 1)$:

$$(K_\sigma \mid K_\sigma)(X) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 \\ \cline{1-2} 1 \\ \cline{1-1} \end{array} \;\middle|\; \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 \\ \cline{1-2} 1 \\ \cline{1-1} \end{array} \right)(X)$$

$$= \text{product of upper-left minors.}$$

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Our goal will be to use $f(X)$ as an oracle to compute a canonical bideterminant.

To do this, we will isolate one of the terms $(S_k \mid T_k)(X)$.

Linear Operators

Definition (Substitution operator)

For $i < j$, the operator $\mathrm{Sub}_{i\to j}$ takes a tableau S and returns the tableau S' formed by taking each row of S which has an i but not a j, replacing that i with j, and sorting the row in increasing order.

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$$\underset{2\to 4}{\mathrm{Sub}} \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 \\ \cline{1-2} 1 \\ \cline{1-1} \end{array} = \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 4 \\ \cline{1-2} 1 \\ \cline{1-1} \end{array}$$

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Let $h_i^j(S)$ be the number of times i is replaced by j in S.

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Let S be a conjugate semistandard tableau of shape σ such that S has entries of value at most n.

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Let $\sigma = (4, 2, 1)$:

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$$(S \mid T)(E_{i,j}(\lambda)X) = \pm\lambda^{h_i^j(S)}(\underset{i\to j}{\mathrm{Sub}}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S' \in \mathcal{C}_{i\to j}^h(S)} \pm(S' \mid T)(X).$$

For $0 \leq h \leq h_i^j(S) - 1$, let $\mathcal{C}_{i\to j}^h(S)$ be the set of tableaux of shape σ obtained by changing i to j at exactly h rows of S which contain i but not j and reordering those rows to be increasing.

Applying all Substitution Operators

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$$M = E_{1,2}(\lambda_{1,2}) \cdots E_{1,n}(\lambda_{1,n}) E_{2,3}(\lambda_{2,3}) \cdots E_{2,n}(\lambda_{2,n}) \cdots E_{n-2,n-1}(\lambda_{n-2,n-1}) E_{n-2,n}(\lambda_{n-2,n}) E_{n-1,n}(\lambda_{n-1,n})$$

$$N = E_{m-1,m}(\xi_{m-1,m})^\top E_{m-2,m}(\xi_{m-2,m})^\top E_{m-2,m-1}(\xi_{m-2,m-1})^\top \cdots E_{2,m}(\xi_{2,m})^\top \cdots E_{2,3}(\xi_{2,3})^\top E_{1,m}(\xi_{1,m})^\top \cdots E_{1,2}(\xi_{1,2})^\top.$$

Applying all Substitution Operators

We have seen now that after applying a sequence of substitution operators $\mathrm{Sub}_{i\to j}$, we can send a bitableau $(S \mid T)$ of shape σ to $(\overline{K}_\sigma \mid \overline{K}_\sigma)$.

Translating this to the setting of linear operators $E_{i,j}(\lambda_{i,j})$ gives the following:

Lemma ([DG25, Lemma 3.9])

Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Then $f(MXN)$ can be expressed as a sum

$$f(MXN) = \sum_{k\in A} \widehat{c}_k \Lambda^{\overline{e}_k} \Xi^{\overline{f}_k} \cdot (\overline{K}_{\sigma_k} \mid \overline{K}_{\sigma_k})(X) + \sum_{\ell\in B} \widehat{c}_\ell \Lambda^{\overline{e}_\ell} \Xi^{\overline{f}_\ell} \cdot (S_\ell \mid T_\ell)(X)$$

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Let J_n be the $n \times n$ matrix with 1's along the anti-diagonal.

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Let J_n be the $n \times n$ matrix with 1's along the anti-diagonal.

Lemma

$$(\overline{K}_\sigma \mid \overline{K}_\sigma)(J_n X J_m) = \pm (K_\sigma \mid K_\sigma)(X).$$

Applying all Substitution Operators

Lemma ([DG25, Corollary 3.10])

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d. Then $f(MXN)$ can be expressed as a sum

$$f(MJ_nXJ_mN) = \sum_{k \in A} \widehat{c}_k \Lambda^{\overline{e}_k} \Xi^{\overline{f}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) + \sum_{\ell \in B} \widehat{c}_\ell \Lambda^{\overline{e}_\ell} \Xi^{\overline{f}_\ell} \cdot (S_\ell \mid T_\ell)(X)$$

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Distinguishing Terms

$$D = \mathrm{diag}(y_1 v, y_2 v^2 \dots, y_n v^n) \in \mathbb{F}[Y, v]^{n \times n}$$
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Lemma

Let $(S \mid T)(X)$ be a bideterminant. Then

$$(S \mid T)(DXD') = Y^{\overline{s}} Z^{\overline{t}} v^{|S|+|T|} \cdot (S \mid T)(X).$$

- $(\overline{s}, \overline{t})$ is the multidegree of $(S \mid T)(X)$: $\mathrm{mdeg}(x_{i,j}) = e_i \oplus e_j$
- $|S|$ is the sum of the entries in S.

Distinguishing Terms

Lemma

Let $f(X) \in I^{\mathrm{det}}_{n,m,r}$ be a nonzero polynomial of degree d. Let $g = f(MJ_nDXD'J_mN) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, v]$.

$$\mathrm{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\overline{e}_k} \Xi^{\overline{f}_k} Y^{\overline{s}_k} Z^{\overline{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X),$$

Linear transformations just correspond to a layer of summation gates giving us an f-oracle circuit for the above.

Use *interpolation* to get an f-oracle circuit computing $\mathrm{coeff}_{v_{d_{min}}}(g)$.

Interpolation

Definition

Consider a degree d polynomial $g(\overline{x}, t) = \sum_{i=0}^{d} \mathrm{coeff}_{t^i}(g) t^i$.

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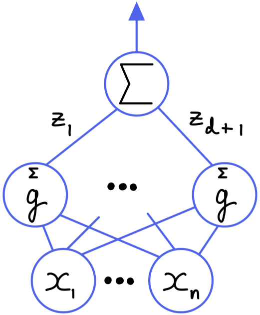

Then, we can compute $\mathrm{coeff}_{t^i}(g)$ by a size $O(ds)$, depth Δ f-oracle circuit.

Proof of Interpolation: Interpolation

Let $\alpha_1, \ldots, \alpha_{d+1}$ be $d+1$ distinct elements in $\mathbb{F}$.

$$\begin{pmatrix} \alpha_1^d & \alpha_1^{d-1} & \cdots & \alpha_1 & 1 \\ \alpha_2^d & \alpha_2^{d-1} & \cdots & \alpha_2 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_d^d & \alpha_d^{d-1} & \cdots & \alpha_d & 1 \\ \alpha_{d+1}^d & \alpha_{d+1}^{d-1} & \cdots & \alpha_{d+1} & 1 \end{pmatrix} \begin{pmatrix} \mathrm{coeff}_{t^d}(f) \\ \mathrm{coeff}_{t^{d-1}}(f) \\ \vdots \\ \mathrm{coeff}_{t}(f) \\ \mathrm{coeff}_{1}(f) \end{pmatrix} = \begin{pmatrix} f(\overline{x}, \alpha_1) \\ f(\overline{x}, \alpha_2) \\ \vdots \\ f(\overline{x}, \alpha_d) \\ f(\overline{x}, \alpha_{d+1}) \end{pmatrix}$$

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This left matrix is a *Vandermonde* matrix and is invertible since the α_i are all distinct.

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$$\mathrm{coeff}_{t^i}(f) = \sum_{j=1}^{d+1} z_{d+1-i,j} f(\overline{x}, \alpha_j).$$

Where Does Isolation Arise?

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We now have an f-oracle circuit to compute

$$g(X) = \sum_{k \in I} c_k \Lambda^{\overline{e}_k} \Xi^{\overline{f}_k} Y^{\overline{s}_k} Z^{\overline{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) \in \mathbb{F}[\Lambda, \Xi, Y, Z][X].$$

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Idea: Interpolate with respect to $\Lambda \cup \Xi \cup Y \cup Z$.

Problem: Interpolating with respect to multiple variables is *expensive*.

Issues with Interpolation

Setup: We want to *isolate* some coefficient $c_{\overline{e}}$ in $g(\overline{x}) = \sum_{\overline{e} \in \mathcal{L}} c_{\overline{e}} \overline{x}^{\overline{e}}$.

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Recall: Prior work [AF22] gave ideal closure results in the border setting.

If a circuit border computes $g(\overline{x})$, then it computes

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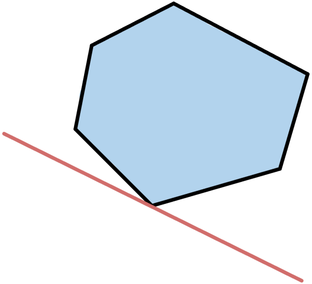
Lemma ([KS01, Lemma 4])

Linear programs with *random* cost functions will have a unique minimum.



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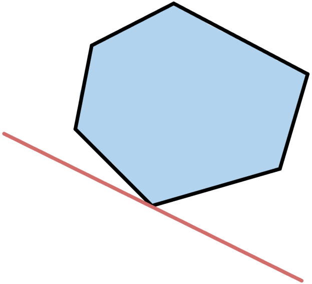

Moreover, if the linear equations have bounded integer coefficients, then evaluation at *small, random* values has a unique minimum.

Isolation Lemma

Lemma ([KS01, Lemma 4])

Let $\mathcal{L}$ be any collection of distinct linear forms in variables $z_1, \ldots, z_\ell$ with coefficients in the range $\{ 0, \ldots, K \}$ for some integer $K \in \mathbb{Z}_{\geq 0}$. Let $\varepsilon > 0$.

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Then, with probability at least $1 - \varepsilon$, there is a unique form in $\mathcal{L}$ of minimum value at $z_1, \ldots, z_\ell$.

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $g(x_1, \ldots, x_\ell)$ such that the individual degree of each x_i in g is at most K:

$$g(x_1, \ldots, x_\ell) = \sum_{\overline{e} \in \mathcal{L}} c_{\overline{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

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The Isolation Lemma shows that the $z_1, \ldots, z_\ell$ can be chosen to be small $\implies$ $\exists$ unique lowest $\deg_w$-term and interpolation w.r.t w is small.

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Theorem ([DG25, Theorem 1.6])

Let $f \in I_{2n,2r}^{\mathrm{pfaff}}$ be a nonzero polynomial. Then there is a depth-three f-oracle circuit of size $\mathrm{poly}(n, \deg(f))$ that *exactly computes* the $t \times t$ Pfaffian for some $t = \mathrm{poly}\left(r^{1/3}\right)$.

Symmetric Analogue?

Let $x_{i,j}$ be indeterminates for $1 \leq i \leq j \leq n$ and let $x_{j,i} \coloneqq x_{i,j}$ for $1 \leq j < i \leq n$. Thus, $X = (x_{i,j})_{1 \leq i,j \leq n}$ is a symmetric $n \times n$ matrix.

Conjecture

Let $I_{n,r}^{\mathrm{sym}} \subseteq \mathbb{F}[X]$ be the ideal generated by the $r \times r$ minors of X. Let $f \in I_{n,r}^{\mathrm{sym}}$ be a nonzero polynomial. Then there exists a constant-depth f-oracle circuit of size $\mathrm{poly}(n, \deg(f))$ that computes the $t \times t$ determinant of a symmetric matrix for $t = \mathrm{poly}(n)$.

Symmetric Basis

Definition (d-tableau, d-determinant)

Following [Con94], we say a bitableau $(S \mid T)$ of shape σ and its associated bideterminant $(S \mid T)(X)$ are *d-tableau* and *d-bideterminants* respectively if S and T are both conjugate semistandard Young tableau and if furthermore the single tableau formed by stacking alternating rows of S and T,

$$\begin{array}{|c|}\hline S_1 \\ \hline T_1 \\ \hline \vdots \\ \hline \vdots \\ \hline S_k \\ \hline T_k \\ \hline \end{array}$$

is conjugate semistandard.

Straightening for Symmetric Determinantal Ideals

Theorem ([dCP76, Theorem 5.1], [Con94, Theorem 2.2])

Let $(S \mid T)$ be a bitableau of shape σ. Then $(S \mid T)(X)$ can be uniquely expressed as a linear combination

$$(S \mid T)(X) = \sum_{(A|B)} c_{A,B}(A \mid B)(X),$$

where the $(A \mid B)(X)$ are d-bideterminants of shape $\tau \geq \sigma$, and $c_{A,B} \in \mathbb{Z}$.

Partial Progress

Theorem

Let $f(X) \in I_{n,r}^{\text{sym}}$ be a nonzero polynomial, d the total degree of f, and $f(X) = \sum_{k\in[s]} \alpha_k (S_k \mid T_k)(X)$ the expansion of $f(X)$ in the d-bideterminant basis where $(S_k \mid T_k)$ is of shape σ_k and each $\alpha_k \in \mathbb{F}$ is nonzero. Then there exists a partition σ and a depth-three f-oracle circuit of size $O\left(d^3 n^7\right)$ computing $\alpha[K_\sigma](X)$ for some nonzero $\alpha \in \mathbb{F}$.

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Our main roadblock for fully resolving the symmetric case is the reduction from a product of determinants to a single determinant.

Roadblocks for the Permanent

The other big star in algebraic complexity is the *permanent* of a matrix.

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Is there an analogue to our results for the ideal of permanents of a matrix?

The main roadblock is that we have no analogue of the following result:

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm\lambda^{h_i^j(S)}(\operatorname*{Sub}_{i\to j}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S' \in \mathcal{C}^h_{i\to j}(S)} \pm(S' \mid T)(X).$$

Algebras with Straightening Laws

Definition ([Eis80])

Let A be a ring, $H \subseteq A$ is a poset. A *standard monomial* is a product $a_1 \cdots a_k$ such that $a_i \in H$ and $a_1 \leq \cdots \leq a_k$.

Let R be a ring, A an R-algebra, and $H \subseteq A$ a finite poset which generates A as an R-algebra. A is an *Algebra with a Straightening Law* if

- A is a free R-module with a basis given by the standard monomials
- if $\alpha, \beta \in H$ are incomparable, and if

$$\alpha\beta = \sum_{i=1}^{m} r_i \gamma_{i,1} \cdots \gamma_{i,k_i}$$

  where $0 \neq r_i \in R$ and each $\gamma_{i,1} \cdots \gamma_{i,k_i}$ is a standard monomial, we must have that for all $1 \leq i \leq m$ that $\gamma_{i,1} \leq \alpha, \beta$.

These relations in the second requirement are known as the *straightening relations* or *straightening laws*.

Hankel Ideals

Let $\mathbb{F}$ be a field. Let x_i be indeterminates for $1 \le i \le n$. For $1 \le j \le n$, let X_j be the following Hankel matrix:

$$\begin{pmatrix} x_1 & x_2 & x_3 & \cdots & x_{n-j+1} \\ x_2 & x_3 & x_4 & \cdots & x_{n-j+2} \\ x_3 & x_4 & x_5 & \cdots & x_{n-j+3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_j & x_{j+1} & x_{j+2} & \cdots & x_n \end{pmatrix}.$$

For $1 \le r \le \min\{\, j, n-j+1 \,\}$ let $I_r(X_j)$ be the ideal generated by $r \times r$ minors of X_j.

Hankel Ideals

The ideals $I_r(X_j)$ do not have the structure of an algebra with a straightening law.

In particular, the product of standard monomials need not be standard.

We still conjecture the following:

Conjecture

Let $I_r(X_j) \subseteq \mathbb{F}[x_1, \ldots, x_n]$ be the ideal generated by the $r \times r$ minors of X_j. Let $f \in I_r(X_j)$ be a nonzero polynomial. Then there exists a constant-depth f-oracle circuit of size $\mathrm{poly}(n, \deg(f))$ that computes the $t \times t$ determinant of a Hankel matrix for $t = \mathrm{poly}(n)$.

Failure of Prior Strategy

In the general determinantal, skew-symmetric, and symmetric cases we analyzed the behavior of $E_{i,j}(\lambda)$.

This no longer works:

$$\begin{pmatrix} 1 & \lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_4 \\ x_3 & x_4 & x_5 \end{pmatrix} = \begin{pmatrix} x_1 + \lambda x_2 & x_2 + \lambda x_3 & x_3 + \lambda x_4 \\ x_2 & x_3 & x_4 \\ x_3 & x_4 & x_5 \end{pmatrix}$$

Thank You!

If you don't know how to answer someone's question, you should complement them by saying it's a good question.

— Noga Alon

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